DAY 29 OF 30

Moving house

Migrating an old warehouse.
Without losing anything on the way.

Assess **Convert** **Reconcile**

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THE PROBLEM

The old warehouse works. That is the problem.

1	**It runs**	But the nightly load takes hours
2	**It is trusted**	But only two people understand it
3	**It is paid for**	But the hardware is ageing

Nobody wants to touch it, and **every year it gets harder to leave**.

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THE SIMPLE VERSION

Moving house without losing anything.

You do not carry everything out in one trip. You list what you own, throw out what you no longer need, move room by room and check nothing is missing.

A migration is the same: **inventory, convert, move, verify.**

✓ **Inventory**	What exists, and what is used
✓ **Move in phases**	One subject area at a time
✓ **Verify**	Old and new must match

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HOW IT WORKS

The phases of a migration

1. **Assess**: tables, procedures, jobs and who uses them
2. **Decide**: move as-is, rewrite, or retire
3. **Convert**: schema and code to the new platform
4. **Run both in parallel** and compare the numbers
5. **Cut over** area by area, then switch the old one off

Parallel running is not optional

Two systems producing the same numbers for a few weeks is how you earn the right to switch.

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FOR ENGINEERS

Tools that help now

Migration Assistant for Data Warehouse

Converts schema and code from Synapse dedicated SQL pools and SQL Server, then moves data with a Copy job.

Spark Migration Assistant

Helps bring Synapse Spark notebooks and pools across.

Data Factory upgrade

Azure Data Factory has an assessment-first upgrade path, and a factory can be mounted in Fabric.

Gateways replace self-hosted runtimes

On-premises connections move to the on-premises data gateway.

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THE HONEST PART

Lift and shift or rewrite?

MOVE AS-IS

Logic is sound and well understood

The deadline is tight

The source is close to T-SQL already

REWRITE

Jobs are slow by design

Logic is duplicated across procedures

Nobody trusts the current output

> **The catch:** a large share of an old warehouse is often unused. Migrating it wastes weeks. Find out what is actually queried before converting a single table.

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FROM REAL WORK

Migrations I have worked on

Two moves away from legacy SQL systems.

1. **An on-premises SQL warehouse** moved to Data Lake and Databricks
2. **Stored procedures** were rewritten as PySpark
3. **Processing time** fell from 2 hr 50 min to about 5 min
4. **A manufacturing group's legacy SQL data** moved to Synapse and Fabric

Stored procedures became **in-memory PySpark**, and the run time followed.

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SAVE THIS

Migration in five lines

1. **Inventory first**, and retire what is unused
2. **Move in phases**, not in one weekend
3. **Rewrite** what was slow by design
4. **Run in parallel** and reconcile
5. **Switch off** the old system on purpose

Tomorrow: The full picture.
One architecture, end to end.

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